EDITABLE

Algebra 2

FINAL EXAM

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Algebra 2: Final Exam

1. ______	**13.** ______	**25.** ______	**37.** ______	**49.** ______
2. ______	**14.** ______	**26.** ______	**38.** ______	**50.** ______
3. ______	**15.** ______	**27.** ______	**39.** ______	**51.** ______
4. ______	**16.** ______	**28.** ______	**40.** ______	**52.** ______
5. ______	**17.** ______	**29.** ______	**41.** ______	**53.** ______
6. ______	**18.** ______	**30.** ______	**42.** ______	**54.** ______
7. ______	**19.** ______	**31.** ______	**43.** ______	**55.** ______
8. ______	**20.** ______	**32.** ______	**44.** ______	**56.** ______
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11. ______	**23.** ______	**35.** ______	**47.** ______	**59.** ______
12. ______	**24.** ______	**36.** ______	**48.** ______	**60.** ______

Check All Answers!

Name: ____

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Algebra 2

Final Exam

SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION! Good Luck! ☺

1. Which expression is equivalent to $\sqrt[3]{324a^{27}b^{8}}$?

A. $3a^3b^2\sqrt[3]{12}$

B. $3a^9b^2\sqrt[3]{12b^2}$

C. $18a^3b^2$

D. $18a^9b^2\sqrt[3]{b^2}$

2. Which expression is equivalent to $\sqrt[4]{8x^7y^3}\cdot\sqrt[4]{32xy^{10}}$?

A. $64x^{\frac{7}{4}}y^{\frac{15}{2}}$

B. $64x^2y^{\frac{13}{4}}$

C. $4x^{\frac{7}{4}}y^{\frac{15}{2}}$

D. $4x^2y^{\frac{13}{4}}$

3. Which expression is equivalent to $2\sqrt{48}-(2+\sqrt{12})(5-\sqrt{12})$?

A. $2\sqrt{3}+2$

B. $2\sqrt{3}-2$

C. $4\sqrt{3}+17$

D. $4\sqrt{3}-17$

4. Which expression is equivalent to $(-\sqrt{8}\cdot\sqrt{2})+\sqrt{-36}$?

A. $2i$

B. $-10i$

C. $-4+6i$

D. $-4-6i$

5. Which expression is equivalent to $-i$?

A. $(i^{15})^3$

B. $(i^6)^7\cdot(i^5)^4$

C. $(1+i)(1-i)$

D. $i^{12}\cdot(i^{13})^3$

6. Which expression is equivalent to $(-7+2i)+6i-(11-15i)$?

A. $-18+23i$

B. $-18-7i$

C. $-4+23i$

D. $-4-7i$

7. Which is a true statement?

A. $m^2-n^2=(m-n)(m-n)$

B. $m^2+n^2=(m+n)(m+n)$

C. $m^3-n^3=(m-n)(m^2+mn+n^2)$

D. $m^3+n^3=(m+n)(m^2+mn+n^2)$

8. Which is the factored form of $64p^3+125$?

A. $(4p+5)^3$

B. $(4p+5)(16p^2+20p+25)$

C. $(4p-5)(16p^2-20p+25)$

D. $(4p+5)(16p^2-20p+25)$

9. Assuming the denominator does not equal zero, simplify the expression below.

$$\frac{6n^2-23n+7}{8n^3-28n^2} \div \frac{1-9n^2}{2n^2-10n}$$

A. $\frac{-2n(n-5)}{3n+1}$

B. $\frac{-(n-5)}{2n(3n+1)}$

C. $\frac{-2n(3n+1)}{n-5}$

D. $\frac{-(3n+1)}{2n(n-5)}$

10. Assuming the denominator does not equal zero, simplify the expression below.

$$\frac{5k}{k-2} - \frac{k}{k+2}$$

A. $\frac{4k(k+3)}{(k+2)(k-2)}$

B. $\frac{4k}{k-2}$

C. $\frac{4k}{(k+2)(k-2)}$

D. $\frac{4k(k-3)}{(k+2)(k-2)}$

11. Assuming no denominator equals zero, simplify the expression below.

$$\frac{\frac{1}{y^2} - \frac{x}{7y^2}}{\frac{1}{y} - \frac{7}{xy}}$$

A. $\frac{-y}{7-x}$

B. $-\frac{7y}{x}$

C. $\frac{-(x-7)}{y}$

D. $-\frac{x}{7y}$

12. What is the solution set for the equation below?

$$2|13-6x|-1=21$$

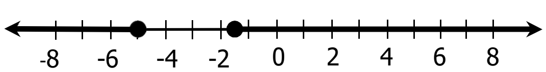
A. $\left\{-4, -\frac{1}{3}\right\}$

B. $\left\{-4, \frac{1}{3}\right\}$

C. $\left\{\frac{1}{3}, 4\right\}$

D. $\left\{-\frac{1}{3}, 4\right\}$

13. Which inequality contains the solutions shown on the graph below?



A. $|4x-13| \geq 7$

B. $|4x-13| \leq 7$

C. $|4x+13| \geq 7$

D. $|4x+13| \leq 7$

14. What is the solution to the following equation?

$$(x-4)^2 = 28?$$

A. $x = 4 \pm 2\sqrt{7}$

B. $x = -4 \pm 2\sqrt{7}$

C. $x = 2$ or $x = 6$

D. $x = -6$ or $x = -2$

15. Which equation has exactly one solution?

A. $3x^2 - 2 = -8$

B. $2x^2 - 14 = 4x - 1$

C. $5x^2 - 4x - 6 = x^2 - 7$

D. $6x^2 = 18 - 23x$

16. What is the solution set to the following equation?

$$4m^2 - 8m + 9 = 0$$

A. $m = 1 \pm 4i\sqrt{5}$

C. $m = -1 \pm 4i\sqrt{5}$

B. $m = \dfrac{2 \pm i\sqrt{5}}{2}$

D. $m = \dfrac{-2 \pm i\sqrt{5}}{2}$

17. What is the solution set to the following equation?

$$x^4 + 2x^2 - 48 = 0$$

A. $\{\pm 2i\sqrt{2}, \pm\sqrt{6}\}$

B. $\{\pm i\sqrt{6}, \pm 2\sqrt{2}\}$

C. $\{\pm 2i\sqrt{2}, \pm 3\}$

D. $\{\pm i\sqrt{3}, \pm 2\sqrt{2}\}$

18. The solutions of the cubic function $f(x)$ are -4, $\frac{5}{3}$, and 1. Identify a factor of this function.

A. $3x + 5$

B. $5x + 3$

C. $x + 1$

D. $x + 4$

19. What is the solution to the following equation?

$$2\sqrt[3]{3x+1} - 11 = -3$$

A. $x = 5$

B. $x = 9$

C. $x = 21$

D. $x = \dfrac{11}{3}$

20. What is the solution to the following equation?

$$\sqrt{a+9} = -3 - a$$

A. $a = 0$

B. $a = -5$

C. $a = \{0, 5\}$

D. $a = \{-5, 0\}$

21. If no denominator is equal to zero, what is the solution set for the following equation?

$$\frac{4v-3}{v^2}=\frac{5}{2v}$$

A. $\{2\}$

B. $\{-2\}$

C. $\left\{-\frac{2}{5},\ 2\right\}$

D. $\left\{-\frac{2}{5},\ \frac{2}{5}\right\}$

22. If no denominator is equal to zero, what is the solution to the following equation?

$$\frac{2k-5}{3k}-\frac{k+2}{4k}=\frac{k+6}{k}$$

A. $k=-14$

B. $k=-\frac{12}{7}$

C. $k=-\frac{5}{12}$

D. $k=-\frac{86}{7}$

23. Which expression is equivalent to $\log\sqrt{a^7b^{16}}$?

A. $\frac{7}{2}\log a+4\log b$

B. $\frac{2}{7}\log a-\frac{1}{4}\log b$

C. $\frac{7}{2}\log a+8\log b$

D. $\frac{2}{7}\log a-\frac{1}{8}\log b$

24. What is the solution to the following equation?

$$\log_2(4c-20)=6$$

A. $c=14$

B. $c=21$

C. $c=8$

D. $c=7$

25. What is the solution to the following equation?

$$2\cdot\log_4(x-1)=\log_4 48-\log_4 3$$

A. $x=\{-3,\ 5\}$

B. $x=\{-5,\ 3\}$

C. $x=3$

D. $x=5$

26. What is the solution to the following equation?

$$9^{2x-15}=\left(\frac{1}{27}\right)^{4x}$$

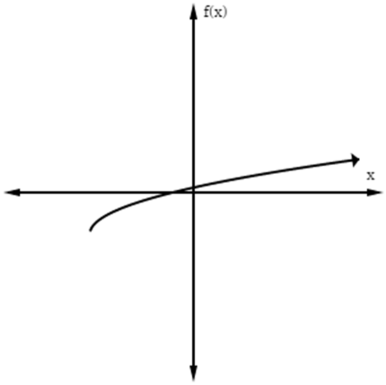
A. $x=-\frac{4}{15}$

B. $x=\frac{8}{15}$

C. $x=-\frac{15}{4}$

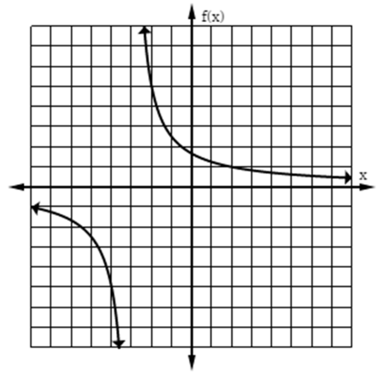
D. $x=\frac{15}{8}$

27. The graph of a function is shown below. Which family could this function belong to?



A. Exponential
B. Logarithmic
C. Square Root
D. Cube Root

28. Which function is best represented by this graph?



A. $f(x) = \dfrac{5}{x+3}$

B. $f(x) = \dfrac{5}{x-3}$

C. $f(x) = \dfrac{x+5}{x+3}$

D. $f(x) = \dfrac{x+5}{x-3}$

29. During which interval is the graph of the function below only increasing?

$$f(x) = x^3 + 9x^2 + 24x + 15$$

A. $-\infty < x < \infty$ **C.** $-5 < x < -2$

B. $-\infty < x < -2$ **D.** $-2 < x < \infty$

30. What is a zero of the function below?

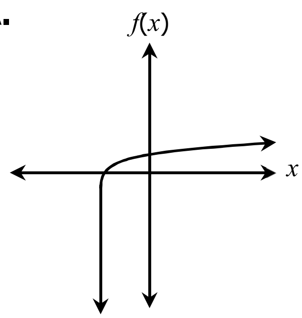
$$f(x) = 3^x - 243$$

A. $x = -5$ **C.** $x = -9$

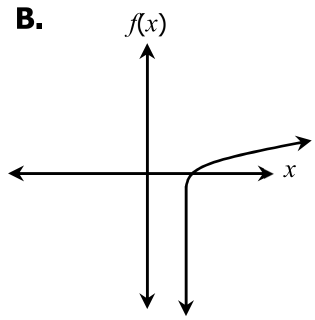
B. $x = 5$ **D.** $x = 9$

31. Which graph could represent the function $f(x) = \log(x - c)$ where $c > 0$?

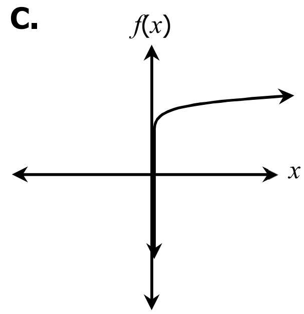
A.



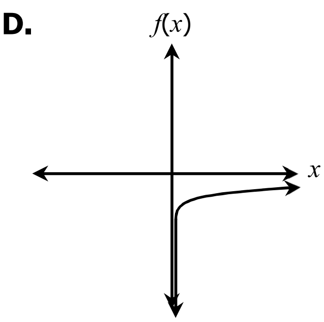
B.



C.



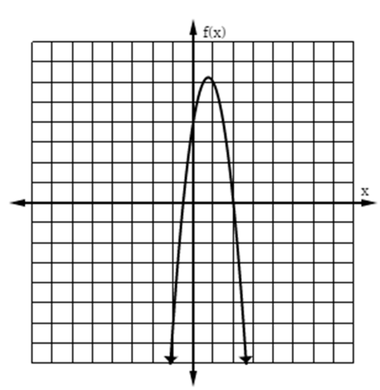
D.



32. Which of the following describes the end behavior of the function $f(x) = \sqrt[3]{x} - 4$ as x approaches infinity?

A. $f(x)$ approaches 0
B. $f(x)$ approaches -4
C. $f(x)$ approaches ∞
D. $f(x)$ approaches $-\infty$

33. Which is an apparent root of the equation graphed below?



A. $\dfrac{1}{2}$

B. $-\dfrac{1}{2}$

C. -2

D. 4

34. Which function does not have a range of the real numbers less than or equal to 0?

A. $f(x) = -(3-x)^2$

B. $f(x) = -|x-3|$

C. $f(x) = -\sqrt{3-x}$

D. $f(x) = \log(x-3)$

35. What is the equation of the horizontal asymptote of the function $f(x) = 3^{x+2} - 7$?

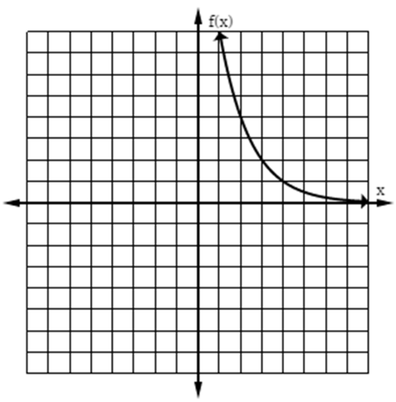
A. $y = 0$

B. $y = 2$

C. $y = 3$

D. $y = -7$

36. Which function best represents this graph?



A. $f(x) = \left(\frac{1}{3}\right)^x + 4$

B. $f(x) = \left(\frac{1}{3}\right)^{x+4}$

C. $f(x) = \left(\frac{1}{3}\right)^x - 4$

D. $f(x) = \left(\frac{1}{3}\right)^{x-4}$

37. Which function has no restrictions in its domain?

A. $f(x) = \log(x-5)$

B. $f(x) = \sqrt{x-5}$

C. $f(x) = \sqrt[3]{x-5}$

D. $f(x) = \frac{1}{x+5}$

38. Comparing the function $f(x) = 2|x-1| + 6$ to its parent function, which transformation did not take place?

A. vertical translation

B. horizontal translation

C. vertical compression

D. vertical stretch

39. Which function decreases through the interval $-\infty < x < 1$ and approaches $-\infty$ as x approaches ∞?

A. $f(x) = x^2 - 2x + 3$

B. $f(x) = -x^2 + 2x - 3$

C. $f(x) = x^3 - 6x^2 + 9x - 6$

D. $f(x) = -x^3 + 6x^2 - 9x + 6$

40. Which gives the vertical and horizontal asymptote of the following function?

$$f(x) = \frac{2x-3}{x+4}$$

A. $x = -4$ and $y = 2$

B. $x = 4$ and $y = 2$

C. $x = 2$ and $y = -4$

D. $x = 2$ and $y = 4$

41. What is the solution set for the following system of equations?

$$\begin{cases} 2x - y = 5 \\ y = 2x^2 - 2x - 11 \end{cases}$$

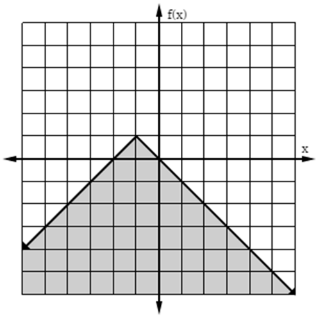
A. $\{\ \}$

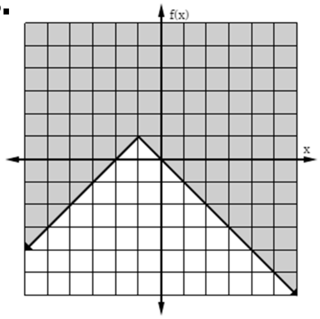
B. $\{(1,3), (7,-1)\}$

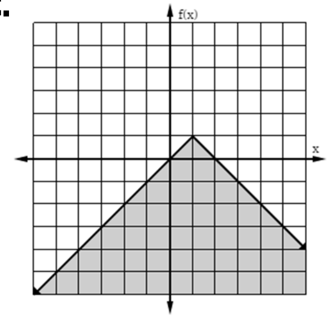
C. $\{(-1,-7), (3,1)\}$

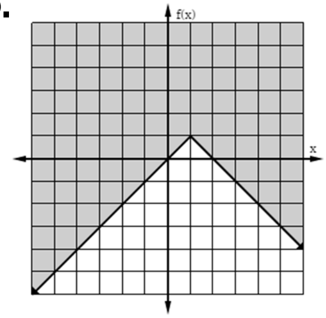
D. $\{(-2, 1), (2,-7)\}$

42. Which function represents the graph $f(x) \le -|x+1|+1$?

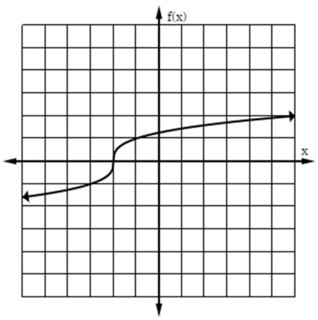
A.


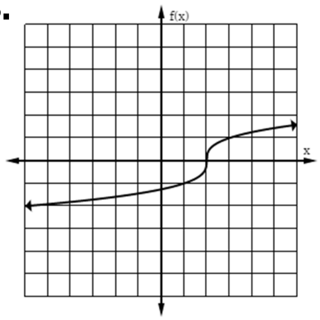
B.


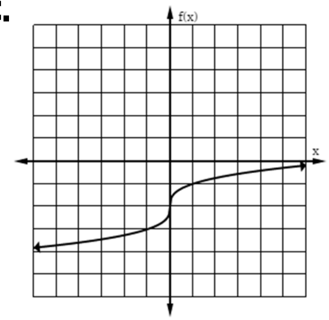
C.


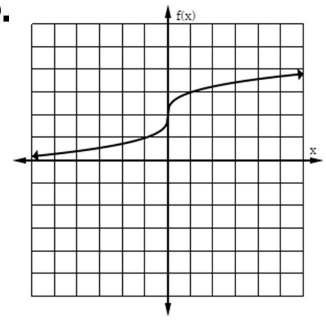
D.


43. Which graph shows the inverse of the function $f(x) = (x+2)^3$?

A.


B.


C.


D.


44. The point (3, 4) lies on the graph of a function. The inverse of this function must contain which point?

A. (-3, -4)
B. (-3, 4)
C. (3, -4)
D. (4, 3)

45. If $f(x) = x^2 + 9x - 14$ and $g(x) = x^2 - x + 3$, find $(f-g)(x)$.

A. $8x - 11$
B. $8x - 17$
C. $10x - 11$
D. $10x - 17$

46. If $f(x) = 3x^2 - 15$ and $g(x) = 1 - \frac{2}{3}x$, find $(f \circ g)(-6)$.

A. 57
B. 60
C. -48
D. -61

47. The work W done when lifting an object varies jointly with the mass m of the object and the square of the height h that the object is lifted. Which equation represents the relationship between work, mass, and volume?

A. $W = kmh^2$
B. $W = \dfrac{k}{mh^2}$
C. $W = k(mh)^2$
D. $W = \dfrac{km}{h^2}$

48. Given: y varies directly as x squared and inversely as z cubed. If y = 12 when x = 4 and z = 2, find x when y = 1.728 and z = 5.

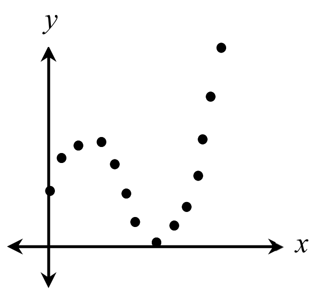
A. $x = 6$
B. $x = 18$
C. $x = 27$
D. $x = 36$

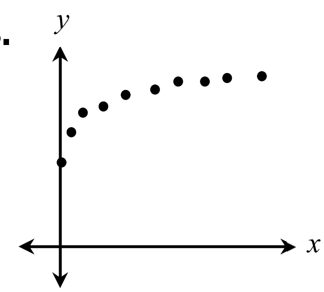
49. The table below shows the population of a city during certain years since 1992. Using an exponential model, estimate the population of the city in 2016.

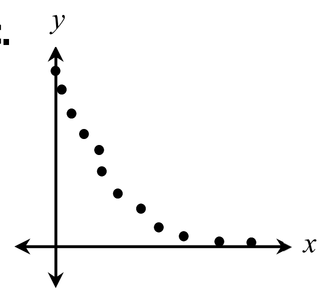
Year	Population
1992	95,242
1995	119,977
1998	151,137
2001	190,389
2004	239,935
2007	302,123

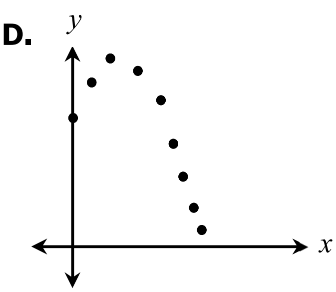
A. 594,164
B. 603,934
C. 617,228
D. 625,572

50. For which set of data would the equation for the curve of best fit most likely be cubic?

A.


B.


C.


D.


51. Given $\begin{cases} a_1 = 2 \\ a_n = 3 \cdot a_{n-1} - 1, \text{ for } n \geq 2 \end{cases}$, find a_5.

A. 41
B. 122
C. 365
D. 2,094

52. What is the 32nd term of an arithmetic sequence with a first term of 7 and a common difference of 4?

A. 119
B. 123
C. 127
D. 131

53. A ball is dropped from a height of 400 feet. Each time it hits the ground, it reaches a height that is three-fifths its previous height. Which formula represents the height of the ball after each bounce?

A. $a_n = 400\left(\frac{3}{5}\right)^{x-1}$

C. $a_n = 400\left(-\frac{3}{5}\right)^{x-1}$

B. $a_n = \frac{3}{5}n + 400$

D. $a_n = -\frac{3}{5}n + 400$

54. Given the series below, find S_{29}.

$$\{13 + 4 + (-5) + (-14) + (-23) + \ldots\}$$

A. -2,480
B. -2,752
C. -3,016
D. -3,277

55. **Evaluate:** $\sum_{i=1}^{8} -3 \cdot (-5)^{i-1}$

A. 8,200
B. 195,312
C. -16,400
D. -292,968

56. **Find the sum of the infinite series below.**

$$\left\{36 - 24 + 16 - \frac{32}{3} + \ldots\right\}$$

A. 108
B. $\frac{108}{5}$
C. $\frac{84}{5}$
D. $\frac{91}{5}$

57. **There are 12 girls and 10 boys in Mrs. Jones's kindergarten class. How many ways can she choose a line leader and caboose each day?**

A. 111
B. 222
C. 231
D. 462

58. **A pizzeria has a choice of 15 toppings. How many ways could someone choose 3?**

A. 15
B. 455
C. 2,730
D. 3,375

59. **The test scores of 1,200 students are normally distributed with a mean of 83 and a standard deviation of 5.5. Under which interval did approximately 978 students score?**

A. $72 < x < 88.5$
B. $77.5 < x < 88.5$
C. $83 < x < 94$
D. $72 < x < 94$

60. **The fitness center held a weight loss competition in which 128 people participated. The amount of weight loss was normally distributed with a mean of 18 pounds and a standard deviation of 5.4 pounds. Approximately how many people lost at least 15 pounds?**

A. 72
B. 78
C. 84
D. 90

Name: ______________________ Date: ________________ Per: ______

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1. B	13. C	25. D	37. C	49. B
2. D	14. A	26. D	38. C	50. A
3. A	15. C	27. C	39. D	51. B
4. C	16. B	28. A	40. A	52. D
5. D	17. A	29. D	41. C	53. A
6. A	18. D	30. B	42. A	54. D
7. C	19. C	31. B	43. C	55. B
8. D	20. B	32. C	44. D	56. B
9. B	21. A	33. B	45. D	57. D
10. A	22. A	34. D	46. B	58. B
11. D	23. C	35. D	47. A	59. A
12. C	24. B	36. D	48. A	60. D

Check All Answers!

Name: ______________________________

Date: __________________________ Bell: ______

Algebra 2

Final Exam

SHOW ALL WORK NEEDED TO ANSWER EACH QUESTION! Good Luck! ☺

1. Which expression is equivalent to $\sqrt[3]{324a^{27}b^8}$?

$\sqrt[3]{27a^{27}b^6}$ $\sqrt[3]{12b^2}$

A. $3a^3b^2\sqrt[3]{12}$

B. $3a^9b^2\sqrt[3]{12b^2}$

C. $18a^3b^2$

D. $18a^9b^2\sqrt[3]{b^2}$

2. Which expression is equivalent to $\sqrt[4]{8x^7y^3} \cdot \sqrt[4]{32xy^{10}}$?

$\sqrt[4]{256x^8y^{13}}$

A. $64x^{\frac{7}{4}}y^{\frac{15}{2}}$

B. $64x^2y^{\frac{13}{4}}$

C. $4x^{\frac{7}{4}}y^{\frac{15}{2}}$

D. $4x^2y^{\frac{13}{4}}$

3. Which expression is equivalent to $2\sqrt{48}-(2+\sqrt{12})(5-\sqrt{12})$?

$8\sqrt{3}-(10+3\sqrt{12}-12)$

$8\sqrt{3}+2-6\sqrt{3}$

$2\sqrt{3}+2$

A. $2\sqrt{3}+2$

B. $2\sqrt{3}-2$

C. $4\sqrt{3}+17$

D. $4\sqrt{3}-17$

4. Which expression is equivalent to $(-\sqrt{8}\cdot\sqrt{2})+\sqrt{-36}$?

$-\sqrt{16}+6i \quad = -4+6i$

A. $2i$

B. $-10i$

C. $-4+6i$

D. $-4-6i$

5. Which expression is equivalent to $-i$?

(i^3)

A. $(i^{15})^3 = i^{45} = i$

B. $(i^6)^7\cdot(i^5)^4 = i^{62} = i^2$

C. $(1+i)(1-i) = 1-i^2 = 2$

D. $i^{12}\cdot(i^{13})^3 = i^{51} = i^3$

6. Which expression is equivalent to $(-7+2i)+6i-(11-15i)$?

$-7+2i+6i-11+15i \quad -18+23i$

A. $-18+23i$

B. $-18-7i$

C. $-4+23i$

D. $-4-7i$

7. Which is a true statement?

A. $m^2-n^2=(m-n)(m-n)$

B. $m^2+n^2=(m+n)(m+n)$

C. $m^3-n^3=(m-n)(m^2+mn+n^2)$

D. $m^3+n^3=(m+n)(m^2+mn+n^2)$

8. Which is the factored form of $64p^3+125$?

$(4p)^3+(5)^3$

A. $(4p+5)^3$

B. $(4p+5)(16p^2+20p+25)$

C. $(4p-5)(16p^2-20p+25)$

D. $(4p+5)(16p^2-20p+25)$

9. Assuming the denominator does not equal zero, simplify the expression below.

$$\frac{6n^2-23n+7}{8n^3-28n^2}\div\frac{1-9n^2}{2n^2-10n}$$

$$\frac{(3n-1)(2n-7)}{4n^2(2n-7)}\cdot\frac{2n(n-5)}{(1+3n)(1-3n)}$$

$$\frac{-(n-5)}{2n(1+3n)}$$

A. $\frac{-2n(n-5)}{3n+1}$ **C.** $\frac{-2n(3n+1)}{n-5}$

B. $\frac{-(n-5)}{2n(3n+1)}$ **D.** $\frac{-(3n+1)}{2n(n-5)}$

10. Assuming the denominator does not equal zero, simplify the expression below.

$$\frac{(k+2)\cdot 5k}{(k+2)(k-2)}-\frac{k\,(k-2)}{(k+2)(k-2)}$$

$$\frac{5k^2+10k}{(k+2)(k-2)}-\frac{k^2-2k}{(k+2)(k-2)}$$

$$\frac{4k^2+12k}{(k+2)(k-2)}$$

A. $\frac{4k(k+3)}{(k+2)(k-2)}$ **C.** $\frac{4k}{(k+2)(k-2)}$

B. $\frac{4k}{k-2}$ **D.** $\frac{4k(k-3)}{(k+2)(k-2)}$

11. Assuming no denominator equals zero, simplify the expression below.

$$\frac{7\cdot\frac{1}{7\cdot y^2}-\frac{x}{7y^2}}{x\cdot\frac{1}{x\cdot y}-\frac{7}{xy}}$$

$$\frac{7-x}{7y^2}\cdot\frac{xy}{x-7}$$

A. $\frac{-y}{7-x}$ **C.** $\frac{-(x-7)}{y}$

B. $-\frac{7y}{x}$ **D.** $-\frac{x}{7y}$

12. What is the solution set for the equation below?

$$2|13-6x|-1=21$$

$13-6x=11$ $13-6x=-11$

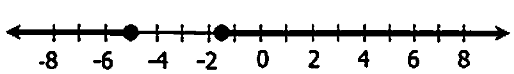
$-6x=-2$ $-6x=-24$

$x=\frac{1}{3}$ $x=4$

A. $\left\{-4,-\frac{1}{3}\right\}$ **C.** $\left\{\frac{1}{3},4\right\}$

B. $\left\{-4,\frac{1}{3}\right\}$ **D.** $\left\{-\frac{1}{3},4\right\}$

13. Which inequality contains the solutions shown on the graph below?



A. $|4x-13|\geq 7$

B. $|4x-13|\leq 7$

C. $|4x+13|\geq 7$

D. $|4x+13|\leq 7$

14. What is the solution to the following equation?

$$(x-4)^2=28?$$

$$x-4=\pm 2\sqrt{7}$$

A. $x=4\pm 2\sqrt{7}$

B. $x=-4\pm 2\sqrt{7}$

C. $x=2$ or $x=6$

D. $x=-6$ or $x=-2$

15. Which equation has exactly one solution?

$[b^2-4ac]$

$4x^2-4x+1$

$(-4)^2-4(4)(1)$

$16-16=0$

A. $3x^2-2=-8$

B. $2x^2-14=4x-1$

C. $5x^2-4x-6=x^2-7$

D. $6x^2=18-23x$

16. What is the solution set to the following equation?

$$4m^2-8m+9=0$$

$x=\dfrac{8\pm\sqrt{(-8)^2-4(4)(9)}}{2(4)}$

$x=\dfrac{8\pm\sqrt{-80}}{8}$ $x=\dfrac{8\pm 4i\sqrt{5}}{8}$

A. $m=1\pm 4i\sqrt{5}$

B. $m=\dfrac{2\pm i\sqrt{5}}{2}$

C. $m=-1\pm 4i\sqrt{5}$

D. $m=\dfrac{-2\pm i\sqrt{5}}{2}$

17. What is the solution set to the following equation?

$$x^4+2x^2-48=0$$

$(x^2+8)(x^2-6)=0$

$x^2=-8$ | $x^2-6=0$

$x=\pm 2i\sqrt{2}$ | $x^2=6$

$x=\pm\sqrt{6}$

A. $\{\pm 2i\sqrt{2},\ \pm\sqrt{6}\}$

B. $\{\pm i\sqrt{6},\ \pm 2\sqrt{2}\}$

C. $\{\pm 2i\sqrt{2},\ \pm 3\}$

D. $\{\pm i\sqrt{3},\ \pm 2\sqrt{2}\}$

18. The solutions of the cubic function $f(x)$ are -4, $\frac{5}{3}$, and 1. Identify a factor of this function.

A. $3x+5$

B. $5x+3$

C. $x+1$

D. $x+4$

19. What is the solution to the following equation?

$$2\sqrt[3]{3x+1}-11=-3$$

$\sqrt[3]{3x+1}=4$

$3x+1=64$

A. $x=5$

B. $x=9$

C. $x=21$

D. $x=\dfrac{11}{3}$

20. What is the solution to the following equation?

$$\sqrt{a+9}=-3-a$$

$a+9=(-3-a)^2$

$a+9=9+6a+a^2$

$0=a^2+5a$

A. $a=0$

B. $a=-5$

C. $a=\{0, 5\}$

D. $a=\{-5, 0\}$

21. If no denominator is equal to zero, what is the solution set for the following equation?

$$\frac{4v-3}{v^2}=\frac{5}{2v}$$

$5v^2 = 8v^2 - 6v$

$-3v^2 = -6v$

$0 = 3v^2 - 6v$

$0 = 3v(v-2)$

A. $\{2\}$

B. $\{-2\}$

C. $\left\{-\frac{2}{5}, 2\right\}$

D. $\left\{-\frac{2}{5}, \frac{2}{5}\right\}$

22. If no denominator is equal to zero, what is the solution to the following equation?

$$\left[\frac{2k-5}{3k}-\frac{k+2}{4k}=\frac{k+6}{k}\right]\cdot 12k$$

$8k - 20 - 3k - 6 = 12k + 72$

$5k - 26 = 12k + 72$

$-98 = 7k$

A. $k=-14$

B. $k=-\frac{12}{7}$

C. $k=-\frac{5}{12}$

D. $k=-\frac{86}{7}$

23. Which expression is equivalent to $\log\sqrt{a^7b^{16}}$?

$\log a^{\frac{7}{2}} b^8$

A. $\frac{7}{2}\log a + 4\log b$

B. $\frac{2}{7}\log a - \frac{1}{4}\log b$

C. $\frac{7}{2}\log a + 8\log b$

D. $\frac{2}{7}\log a - \frac{1}{8}\log b$

24. What is the solution to the following equation?

$$\log_2(4c-20)=6$$

$2^6 = 4c - 20$

$84 = 4c$

$21 = c$

A. $c=14$

B. $c=21$

C. $c=8$

D. $c=7$

25. What is the solution to the following equation?

$$2\cdot\log_4(x-1)=\log_4 48-\log_4 3$$

$(x-1)^2 = 16$

$x^2 - 2x + 1 = 16$

$x^2 - 2x - 15 = 0$

$(x-5)(x+3) = 0$

$x = 5, \cancel{-3}$

A. $x=\{-3, 5\}$

B. $x=\{-5, 3\}$

C. $x=3$

D. $x=5$

26. What is the solution to the following equation?

$$9^{2x-15}=\left(\frac{1}{27}\right)^{4x}$$

$3^{2(2x-15)} = 3^{-3(4x)}$

$4x - 30 = -12x$

$-30 = -16x$

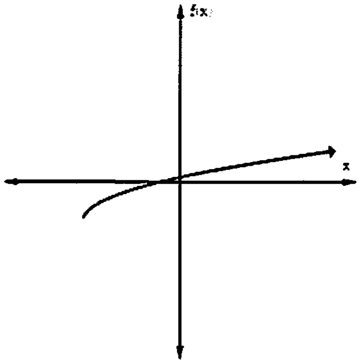
A. $x=-\frac{4}{15}$

B. $x=\frac{8}{15}$

C. $x=-\frac{15}{4}$

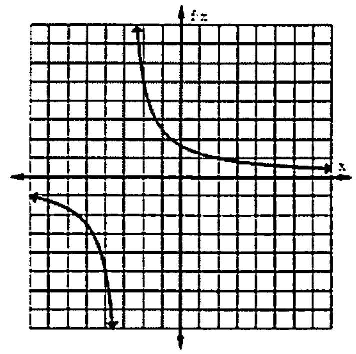
D. $x=\frac{15}{8}$

27. The graph of a function is shown below. Which family could this function belong to?



A. Exponential
B. Logarithmic
C. Square Root
D. Cube Root

28. Which function is best represented by this graph?



A. $f(x)=\dfrac{5}{x+3}$
B. $f(x)=\dfrac{5}{x-3}$
C. $f(x)=\dfrac{x+5}{x+3}$
D. $f(x)=\dfrac{x+5}{x-3}$

29. During which interval is the graph of the function below only increasing?

$$f(x)=x^3+9x^2+24x+15$$

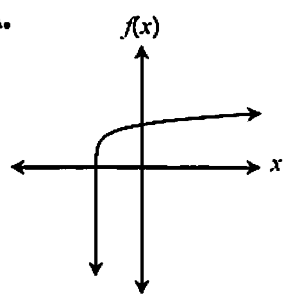
A. $-\infty<x<\infty$
B. $-\infty<x<-2$
C. $-5<x<-2$
D. $-2<x<\infty$

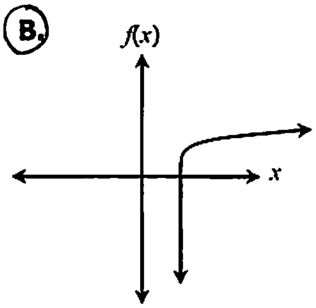
30. What is a zero of the function below?

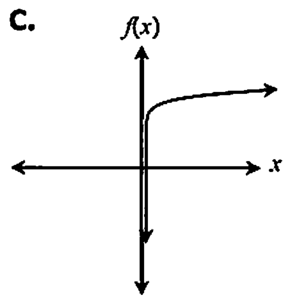
$$f(x)=3^x-243$$

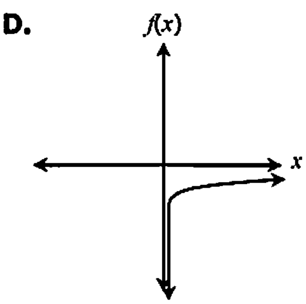
A. $x=-5$
B. $x=5$
C. $x=-9$
D. $x=9$

31. Which graph could represent the function $f(x)=\log(x-c)$ where $c>0$?

A.


B.


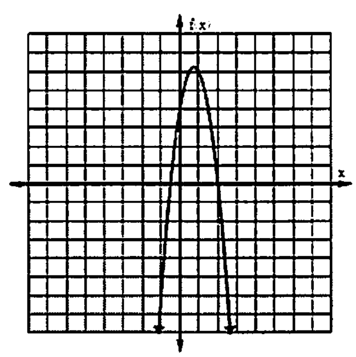
C.


D.


32. Which of the following describes the end behavior of the function $f(x)=\sqrt[3]{x}-4$ as x approaches infinity?

A. $f(x)$ approaches 0
B. $f(x)$ approaches -4
C. $f(x)$ approaches ∞
D. $f(x)$ approaches $-\infty$

33. Which is an apparent root of the equation graphed below?



A. $\frac{1}{2}$
B. $-\frac{1}{2}$
C. -2
D. 4

34. Which function does not have a range of the real numbers less than or equal to 0?

A. $f(x) = -(3-x)^2$

B. $f(x) = -|x-3|$

C. $f(x) = -\sqrt{3-x}$

D. $f(x) = \log(x+3)$

35. What is the equation of the horizontal asymptote of the function $f(x) = 3^{x+2} - 7$?

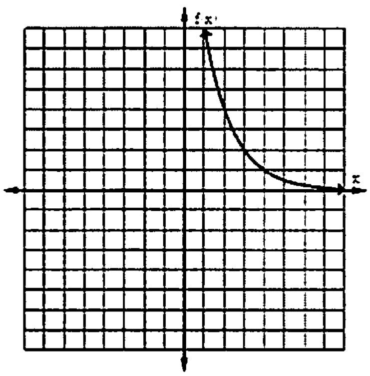
A. $y = 0$

B. $y = 2$

C. $y = 3$

D. $y = -7$

36. Which function best represents this graph?



A. $f(x) = \left(\frac{1}{3}\right)^x + 4$

B. $f(x) = \left(\frac{1}{3}\right)^{x+4}$

C. $f(x) = \left(\frac{1}{3}\right)^x - 4$

D. $f(x) = \left(\frac{1}{3}\right)^{x-4}$

37. Which function has no restrictions in its domain?

A. $f(x) = \log(x-5)$

B. $f(x) = \sqrt{x-5}$

C. $f(x) = \sqrt[3]{x-5}$

D. $f(x) = \frac{1}{x+5}$

38. Comparing the function $f(x) = 2|x-1| + 6$ to its parent function, which transformation did not take place?

A. vertical translation

B. horizontal translation

C. vertical compression

D. vertical stretch

39. Which function decreases through the interval $-\infty < x < 1$ and approaches $-\infty$ as x approaches ∞?

A. $f(x) = x^2 - 2x + 3$

B. $f(x) = -x^2 + 2x - 3$

C. $f(x) = x^3 - 6x^2 + 9x - 6$

D. $f(x) = -x^3 + 6x^2 - 9x + 6$

40. Which gives the vertical and horizontal asymptote of the following function?

$$f(x) = \frac{2x-3}{x+4}$$

A. $x = -4$ and $y = 2$

B. $x = 4$ and $y = 2$

C. $x = 2$ and $y = -4$

D. $x = 2$ and $y = 4$

41. What is the solution set for the following system of equations?

$$\begin{cases} 2x - y = 5 \\ y = 2x^2 - 2x - 11 \end{cases}$$

$2x-5 = 2x^2 - 2x - 11$

$0 = 2x^2 - 4x - 6$

$0 = 2(x^2 - 2x - 3)$

$0 = 2(x-3)(x+1)$

$x = 3$ $\quad$ $x = -1$

$y = 1$ $\quad$ $y = -7$

A. $\{\ \}$

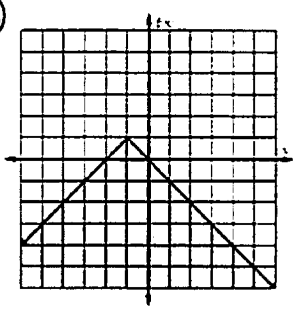
B. $\{(1,3), (7,-1)\}$

C. $\{(-1,-7), (3,1)\}$

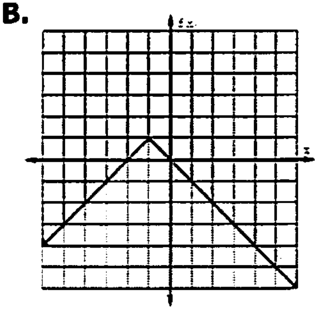
D. $\{(-2,1), (2,-7)\}$

42. Which function represents the graph $f(x) \le -|x+1|+1$?

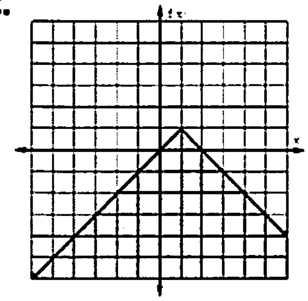
A.



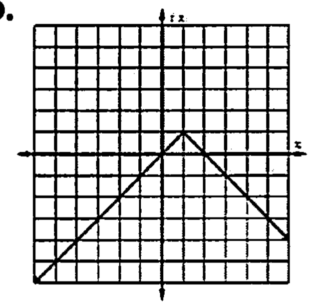
B.



C.

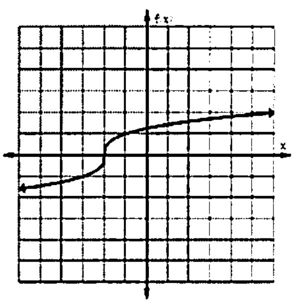


D.

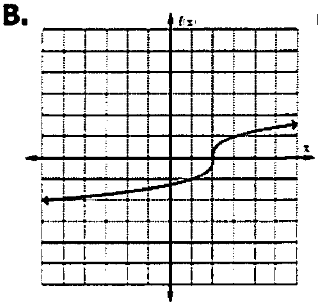


43. Which graph shows the inverse of the function $f(x)=(x+2)^3$?

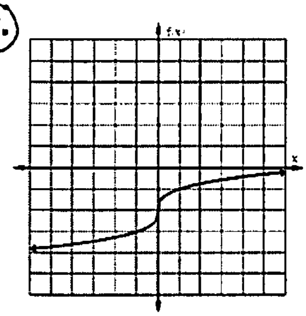
A.



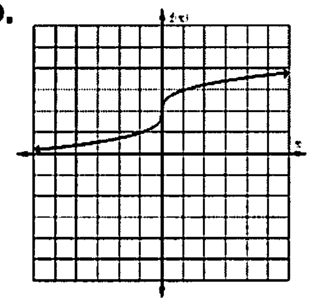
B.



C.



D.



44. The point (3, 4) lies on the graph of a function. The inverse of this function must contain which point?

A. (-3, -4)

B. (-3, 4)

C. (3, -4)

D. (4, 3)

45. If $f(x)=x^2+9x-14$ and $g(x)=x^2-x+3$, find $(f-g)(x)$.

$(x^2+9x-14)-(x^2-x+3)$

A. $8x-11$

B. $8x-17$

C. $10x-11$

D. $10x-17$

46. If $f(x)=3x^2-15$ and $g(x)=1-\frac{2}{3}x$, find $(f \circ g)(-6)$.

$g(-6) = 1-\frac{2}{3}(-6) = 5$

$f(5) = 3(5)^2-15 = 60$

A. 57

B. 60

C. -48

D. -61

47. The work W done when lifting an object varies jointly with the mass m of the object and the square of the height h that the object is lifted. Which equation represents the relationship between work, mass, and volume?

A. $W = kmh^2$

B. $W = \dfrac{k}{mh^2}$

C. $W = k(mh)^2$

D. $W = \dfrac{km}{h^2}$

48. Given: y varies directly as x squared and inversely as z cubed. If $y = 12$ when $x = 4$ and $z = 2$, find x when $y = 1.728$ and $z = 5$.

$y = \frac{kx^2}{z^3}$ $\quad 12 = \frac{k(4)^2}{(2)^3}$ $\quad k=6$

$1.728 = \frac{6x^2}{5^3}$

$x^2 = 36$

A. $x = 6$
B. $x = 18$
C. $x = 27$
D. $x = 36$

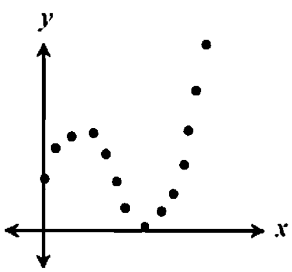
49. The table below shows the population of a city during certain years since 1992. Using an exponential model, estiamte the population of the city in 2016. $(x=24)$

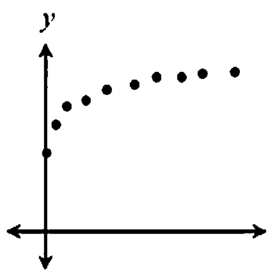
$y = 95239 \cdot 1.08^x$

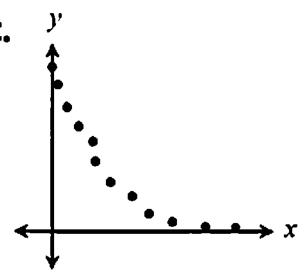
	Year	Population
0	1992	95,242
3	1995	119,977
6	1998	151,137
9	2001	190,389
12	2004	239,935
15	2007	302,123

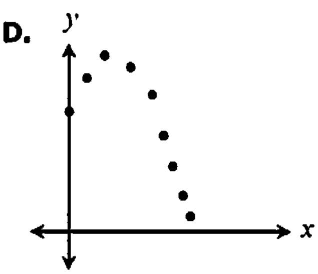
A. 594,164
B. 603,934
C. 617,228
D. 625,572

50. For which set of data would the equation for the curve of best fit most likely be cubic?

A.


B.


C.


D.


51. Given $\begin{cases} a_1 = 2 \\ a_n = 3 \cdot a_{n-1} - 1, \text{ for } n \geq 2 \end{cases}$, find a_5.

2, 5, 14, 41, 122

A. 41
B. 122
C. 365
D. 2,094

52. What is the 32nd term of an arithmetic sequence with a first term of 7 and a common difference of 4?

$4(32-1)+7$

A. 119
B. 123
C. 127
D. 131

53. A ball is dropped from a height of 400 feet. Each time it hits the ground, it reaches a height that is three-fifths its previous height. Which formula represents the height of the ball after each bounce?

A. $a_n = 400\left(\frac{3}{5}\right)^{x-1}$
B. $a_n = \frac{3}{5}n + 400$
C. $a_n = 400\left(-\frac{3}{5}\right)^{x-1}$
D. $a_n = -\frac{3}{5}n + 400$

54. Given the series below, find S_{29}.

$a_{29} = -9(29-1)+13$

$$\{13 + 4 + (-5) + (-14) + (-23) + \ldots\}$$

$S_n = 29\left(\frac{13 + -239}{2}\right)$

A. -2,480
B. -2,752
C. -3,016
D. -3,277

55. Evaluate: $\sum_{i=1}^{8} -3 \cdot (-5)^{i-1}$

$S_8 = \frac{-3(1-(-5)^8)}{1-(-5)}$

A. 8,200
B. 195,312
C. -16,400
D. -292,968

56. Find the sum of the infinite series below.

$$\left\{36 - 24 + 16 - \frac{32}{3} + \ldots\right\}$$

$S = \frac{36}{1-(-2/3)}$

A. 108
B. $\frac{108}{5}$
C. $\frac{84}{5}$
D. $\frac{91}{5}$

57. There are 12 girls and 10 boys in Mrs. Jones's kindergarten class. How many ways can she choose a line leader and caboose each day?

${}_{22}P_2$

A. 111
B. 222
C. 231
D. 462

58. A pizzeria has a choice of 15 toppings. How many ways could someone choose 3?

${}_{15}C_3$

A. 15
B. 455
C. 2,730
D. 3,375

59. The test scores of 1,200 students are normally distributed with a mean of 83 and a standard deviation of 5.5. Under which interval did approximately 978 students score?

$P(-2,1) = .815 \rightarrow .815(1200)$
$= 978$ students

A. $72 < x < 88.5$
B. $77.5 < x < 88.5$
C. $83 < x < 94$
D. $72 < x < 94$

60. The fitness center held a weight loss competition in which 128 people participated. The amount of weight loss was normally distributed with a mean of 18 pounds and a standard deviation of 5.4 pounds. Approximately how many people lost at least 15 pounds?

A. 72
B. 78
C. 84
D. 90

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